

A Rapid Class-Shape-Transformation and Viscous–Potential Framework for Engine-Nacelle Drag Optimization

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Abstract

The nacelle lip controls capture area, mass flow and external drag, but evaluating it well usually demands expensive Reynolds-averaged Navier–Stokes (RANS) solves. This report presents a fully integrated Python framework that couples a Class Shape Transformation (CST) parameterization of the lip and cowl to an in-house viscous–potential solver, so that high-fidelity topological transitions of the geometry can be screened at a fraction of the cost. The stream function is solved by finite differences for local Mach and pressure fields, and a boundary-layer Reynolds-number model supplies empirical skin friction, so that form and friction drag are resolved separately and coupled afterwards. Sweeping the lip flare angle exposes the mass-flow-versus-drag trade and localizes drag minima and incipient separation, allowing rapid lip morphing to deliver form-drag reductions as a design filter ahead of RANS.

Keywords: nacelle; Class Shape Transformation; viscous–potential method; lip morphing; drag optimization; conceptual design

I Introduction

The geometry of an engine nacelle sets three coupled quantities at once: the capture area of the inlet streamtube, the mass flow delivered to the engine, and a large part of the external drag. It also changes with the propulsion layout, because the cowl and lip sit differently for a podded engine, a circulation-control wing with embedded fans, or a distributed blown wing. Evaluating these trade-offs well normally means high-fidelity RANS solves, whose geometric and temporal cost makes design iteration slow; panel and vortex-lattice methods are fast but miss the coupled internal–external flow that the lip produces.

This report presents a framework that closes that gap. A CST parameterization gives dynamic, localized control of the lip and cowl, and an in-house viscous–potential solver evaluates each candidate geometry cheaply while still accounting for compressibility and a viscous drag penalty. Used as a screen ahead of RANS, the method identifies localized drag minima and separation points and acts as a superior design filter in the conceptual aerospace design loop.

II Geometric parameterization

The nacelle cross-section (Fig. 1) is described with a Class Shape Transformation [1]. The radial thickness distribution of the cowl, $\Delta R(\psi)$, is written along the normalized longitudinal station $\psi \in [0, 1]$ as the product of a class function and a shape function,

$$\Delta R(\psi) = C(\psi)S(\psi), \quad C(\psi) = \psi^{N_1}(1 - \psi)^{N_2}, \quad (1)$$

where the exponents (N_1, N_2) fix the class of shape (a rounded inlet lip with a sharp or open exhaust), and the shape function is a Bernstein-polynomial expansion,

$$S(\psi) = \sum_{i=0}^n b_i \binom{n}{i} \psi^i (1 - \psi)^{n-i}. \quad (2)$$

The control coefficients b_i give independent, localized control of the upper and lower lip lobes and the cowl, enabling rapid generation of nacelles that satisfy engine packaging constraints. The two parameters the optimizer pivots are the lip flare angle and the contraction ratio: a sharply inward lip detaches the stagnation streamline and produces severe suction peaks and pressure drag, while a prominent bell-mouth outward flare aligns the flow but penalizes frontal area and skin friction. Neither extreme is efficient, which is exactly why a fast sweep of the trade is valuable.

III Rapid aerodynamic evaluation

The mass flow ingested by the engine must satisfy continuity,

$$\dot{m} = \rho AV, \quad (3)$$

and the capture streamtube expands or contracts with flight speed. To capture compressibility without a full RANS solve, the parameterized nacelle is exported to a two-dimensional fluid domain and the stream function is solved by finite differences, yielding the local Mach number and pressure fields instantaneously (Figs. 2, 3). Viscous penalties are added through a boundary-layer approximation: a local Reynolds number

$$Re_x = \frac{\rho V x}{\mu} \quad (4)$$

drives an empirical skin-friction coefficient $C_f(Re_x)$ over the wetted surface. The form drag and the friction drag are computed separately and coupled afterwards,

$$C_D = C_{D,\text{form}} + C_{D,\text{friction}}, \quad (5)$$

which produces a nuanced drag profile and lets the framework localize both drag minima and incipient separation.

IV Results

Sweeping the lip flare angle while tracking the total drag coefficient and the ingested mass-flow rate exposes the core design trade directly (Fig. 4). The mass flow rises smoothly and almost monotonically as the lip flares outward, while the drag coefficient is markedly non-monotonic, with localized minima that a brute-force optimizer would mis-map. Screening candidate geometries on this map before committing to a Navier–Stokes solver, and applying rapid localized lip morphing around the identified minima, yields significant form-drag reductions while keeping the design loop fast enough

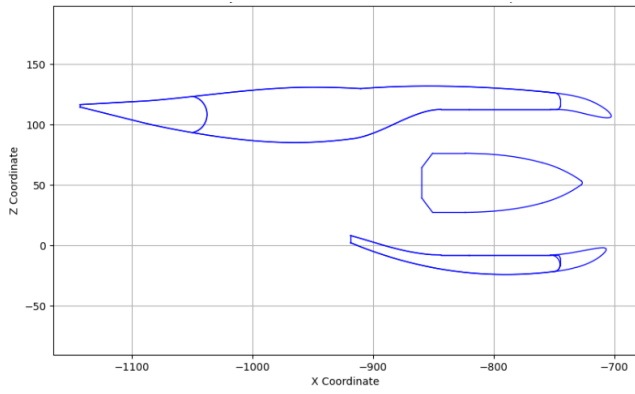


Figure 1. Nacelle cross-section: cowl, lip and centrebody, parameterized by Class Shape Transformation.

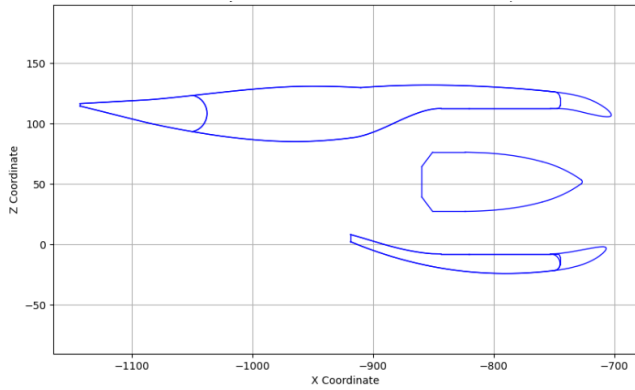


Figure 2. CST representation of the cowl, allowing the lip and cowl to be morphed continuously.

for interactive use. A throat and lip map of the kind used to filter geometries is shown in Fig. 5.

V Conclusion

A CST parameterization coupled to a viscous–potential solver turns nacelle lip design from an expensive RANS campaign into an interactive screening problem. By resolving form and friction drag separately and localizing drag minima and separation, the framework serves as a superior design filter ahead of high-fidelity analysis, with reduced mathematical complexity and computational cost.

References

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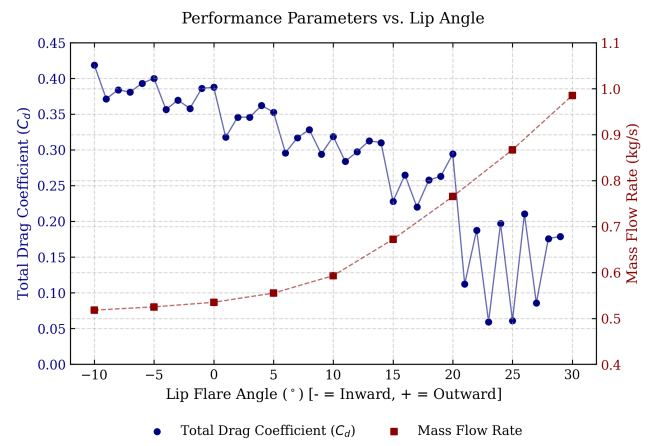


Figure 4. Total drag coefficient and ingested mass-flow rate traced against lip flare angle, the central design trade.

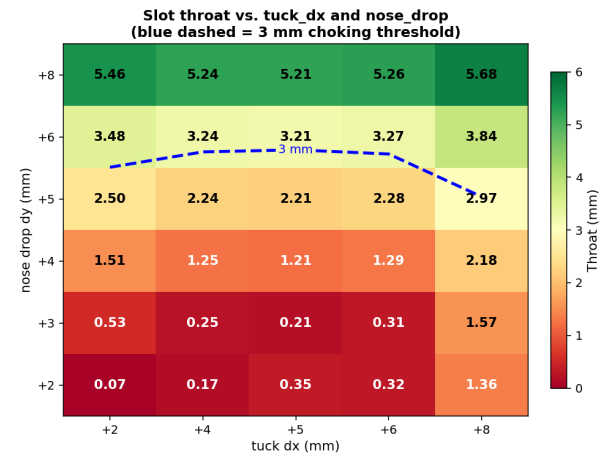


Figure 5. Throat and lip design map used to filter geometries before high-fidelity runs.

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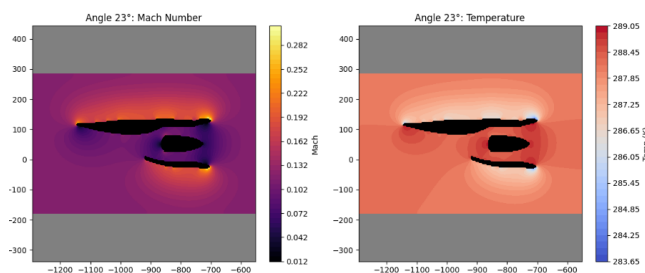


Figure 3. Representative flow solution from the viscous–potential solver used to screen each geometry.